

Online Retail II: Transaction Anomaly Review

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1. Executive Summary

Objective. To apply layered analytical procedures to the client's sales ledger to identify transactions exhibiting characteristics that warrant further audit enquiry, and to evaluate the reliability of those procedures against a controlled, labelled validation set.

Scope. The population comprises 1,013,930 transaction lines (£37,645,678 total value) covering December 2009 to December 2011. 19,494 cancellation entries and non-product ledger adjustments were identified and excluded from digit-distribution and model-based testing, but are retained in the dataset and counted separately; they are not deleted from the population under review.

Procedures performed. (i) Digit-distribution analysis (Benford's Law), aggregate and segmented by country and calendar month; (ii) ten deterministic rule-based exception tests (duplicate invoices, round amounts, threshold structuring, off-hours and period-end posting, among others); (iii) an unsupervised statistical outlier procedure (Isolation Forest) blended with (i) and (ii) into a single composite risk score.

Summary of findings

#	Procedure	Result	Verdict
1	Aggregate Benford (leading digit)	MAD 0.0251	NONCONFORMING
2	Segmented Benford (country x month)	66 / 66 segments flagged	See sec. 3
3	Rule-based exception testing	194,792 / 1,030,804 lines flagged (18.9%)	See sec. 4
4	Isolation Forest (statistical outliers)	Average Precision 0.136	See sec. 5
5	Composite risk ranking	15008 items scored CRITICAL	See sec. 5-6

Conclusion. Analytical procedures identified 207,899 transactions (20.2% of the 1,030,804-transaction scored population, drawn from a representative sample of the full ledger) exhibiting characteristics consistent with the risk indicators tested. These results are indicators for targeted testing and do not, of themselves, evidence misstatement or fraud. A synthetic validation exercise (sec. 2) indicates the composite ranking concentrates known anomalies well above the rate expected from random selection, but every precision figure quoted in this workpaper is a lower bound; see the limitations in sec. 2.

2. Scope, Data and Limitations

Population reconciliation

Stage	Rows
Raw combined rows	1,067,371
Less: cancellations	(19,494)
Less: exact duplicates	(33,947)
Cleaned population (rows_out)	1,013,930
of which: adjustments (retained, excluded from Benford)	4,580
of which: non-positive amount (retained, excluded from Benford)	6,037
of which: missing customer_id (retained as UNASSIGNED)	243,007

Reconciliation holds: raw rows less exclusions equals the cleaned population (reconciles = True). No row was silently dropped; every exclusion is counted above and retained in the underlying dataset for inspection.

Data reliability

The population is a real, publicly available transaction-level extract (UCI Machine Learning Repository, "Online Retail II", CC BY 4.0). It was not generated for this exercise and required only standard cleaning (type coercion, cancellation and duplicate identification) prior to testing.

Synthetic anomaly disclosure

A controlled set of 16,874 synthetic anomalies (1.64% of the scored population) was deliberately injected into the real ledger prior to testing, across six archetypes (duplicate billing, threshold structuring, round-number entries, digit fabrication, cut-off / off-hours timing, and gross-value outliers). This was done because live engagement data carries no ground truth against which a procedure's precision and recall can be measured; the injected set provides that ground truth for this validation exercise only and has no bearing on any conclusion about the real, un-injected transactions in the underlying ledger.

Limitations

- Positives in the validation exercise are the injected rows only; a genuine, unlabelled anomaly already present in the real ledger is scored as a false positive when a procedure flags it, so every precision figure in this workpaper is a lower bound.
- The Isolation Forest decision threshold was calibrated to the known injection rate; a real engagement does not know the true anomaly rate in advance.
- The statistical outlier procedure was evaluated on a 50,000-line sample of the 1,013,930-line population, not the full ledger.
- This is a retail sales ledger, not a general ledger with journal entries; certain indicators (posting user, manual-versus-automatic source, account combinations) have no equivalent in this data.
- A Benford exception is a screening result. It directs procedures; it does not, on its own, indicate fraud or misstatement.

3. Digit-Distribution Analysis (Benford's Law)

Method. Benford's Law states that the leading digit of many naturally occurring numeric populations follows $\log_{10}(1 + 1/d)$, not a uniform distribution. Conformity is assessed primarily via **Mean Absolute Deviation (MAD)**, the mean absolute difference between observed and expected digit proportions, using Nigrini's thresholds (Close <0.006, Acceptable <0.012, Marginal <0.015, Nonconformity ≥ 0.015 ; Nigrini, *Benford's Law: Applications for Forensic Accounting, Auditing, and Fraud Detection*, Wiley, 2012).

Aggregate result. Population $n = 990,028$; $MAD = 0.0251$; verdict **NONCONFORMING**. A chi-square statistic was also computed ($\chi^2 = 990,026$, $p < 0.001$) but is not the primary criterion: at this population size chi-square rejects conformity for deviations of no practical significance (the well-documented "excess power" property of the test at large n). MAD, which is sample-size independent, governs the verdict.

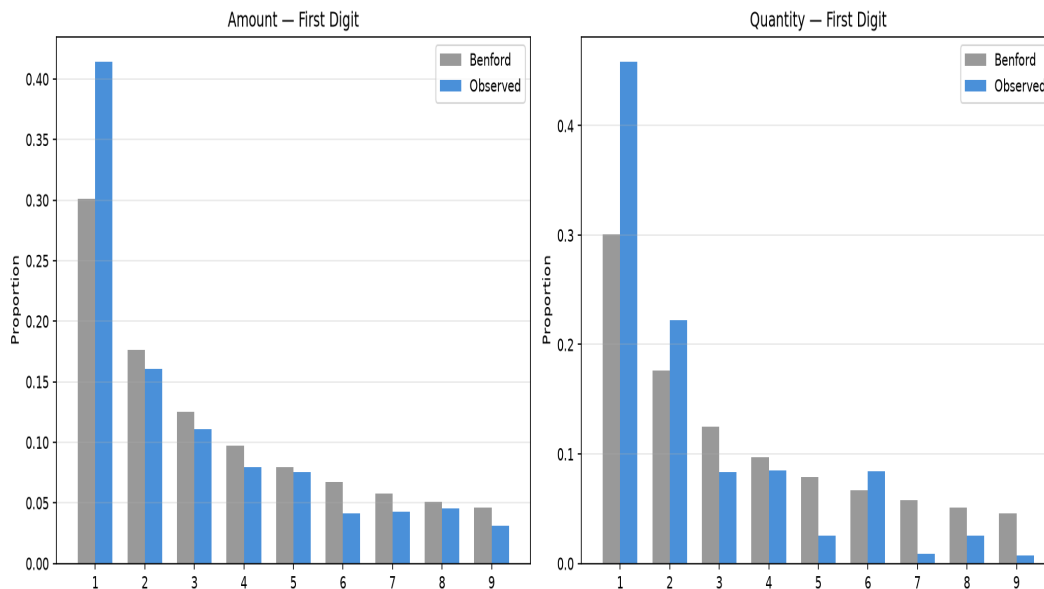


Figure 1: Leading-digit distribution, aggregate population, observed vs. expected.

Segment results. 66 (country, calendar-month) segments met the minimum size for a conformity verdict ($n \geq 1,000$); 66 were classified NONCONFORMING. The five most nonconforming segments:

Country month	n	MAD	Verdict	Digit driving deviation
Eire 2011-09	1,211	0.0668	NONCONFORMING	1
Eire 2011-11	1,097	0.0636	NONCONFORMING	1
Eire 2010-11	1,121	0.0626	NONCONFORMING	1
Germany 2011-10	1,321	0.0613	NONCONFORMING	1
France 2011-09	1,061	0.0573	NONCONFORMING	1

Finding. Every segment large enough to test was classified nonconforming in this review. This is itself a finding worth escalating (extend testing across the population rather than a targeted subset), though it also means the segment flag alone did not differentiate individual transactions for review in this exercise (see sec. 5).

4. Rule-Based Exception Testing

Ten deterministic tests were applied, each with a stated audit rationale. Thresholds and exception counts:

Test	Audit rationale	Exceptions
ROUND_NUMBER	Amount is a round multiple (e.g. 1,000): manual-entry indicator	113,491
SPLIT_AMOUNT	Amount just below an authorisation threshold: structuring risk	68,925
ROUND_DOLLARS	Amount has zero pence: manual-entry indicator	28,106
DUPLICATE_INVOICE	Same customer/amount/date, different invoice: duplicate billing risk	22,443
LARGE_AMOUNT	Amount above a fixed high-value threshold: requires senior review	10,300
HIGH_QUANTITY	Quantity far above the customer's typical order size	9,979
NEGATIVE_ADJUSTMENT	Negative amount outside the normal returns pattern	5
ZERO_QUANTITY	Zero-quantity line with a non-zero amount	0
ODD_QUANTITY	Quantity pattern inconsistent with normal ordering	0
OFF_HOUR	Posted outside business hours (07:00-19:59)	0

Control observation. 194,792 of 1,030,804 lines (18.9%) carried at least one exception flag; 888 lines carried four or more, indicating a small population of lines warranting priority review on multiple independent criteria.

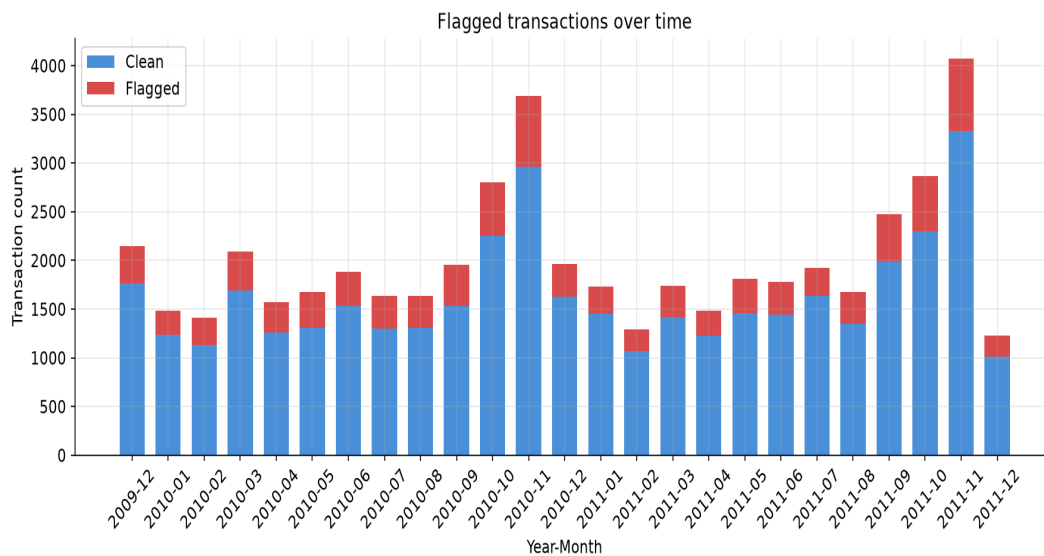


Figure 2: Rule-flag activity over time.

5. Statistical Outlier Analysis and Combined Results

An unsupervised algorithm (Isolation Forest) was used to score how easily each transaction can be separated from the rest of the population by its numeric attributes (amount, quantity, price, time of day, and the customer's historic transaction profile), transactions that separate in fewer steps receive a higher score.

Validation results. Against the labelled synthetic set, the composite score achieved an Average Precision of 0.166 (95% confidence interval 0.160-0.171) against a 0.016 baseline expected from random selection, a lower bound on real-world precision, for the reasons given in sec. 2. This supports treating the composite ranking as a reasonable basis for prioritising review effort; it does not, on its own, support any conclusion about individual transactions without further enquiry.

Type	benford_any	composite	iforest	lof	rules_any
digit_fabrication	93.7%	1.2%	2.5%	2.5%	45.8%
duplicate	90.7%	2.5%	2.9%	0.2%	25.3%
extreme_outlier	91.3%	37.2%	24.5%	3.9%	84.8%
round_number	93.0%	70.6%	71.7%	1.5%	100.0%
threshold_avoidance	92.8%	52.7%	55.6%	1.7%	100.0%
timing	92.7%	0.9%	1.1%	0.3%	16.0%
ALL_INJECTED	92.3%	28.4%	27.6%	1.6%	63.4%
REAL_ROWS	92.9%	1.1%	1.1%	0.2%	18.2%

Interpretation. No single procedure dominated every category in this review: rule-based tests caught the highest share of every injected category at a matched review budget, while the segmented Benford test surfaced fabricated-digit patterns that the aggregate test alone did not reveal. This is consistent with the case for applying more than one procedure rather than relying on any single test.

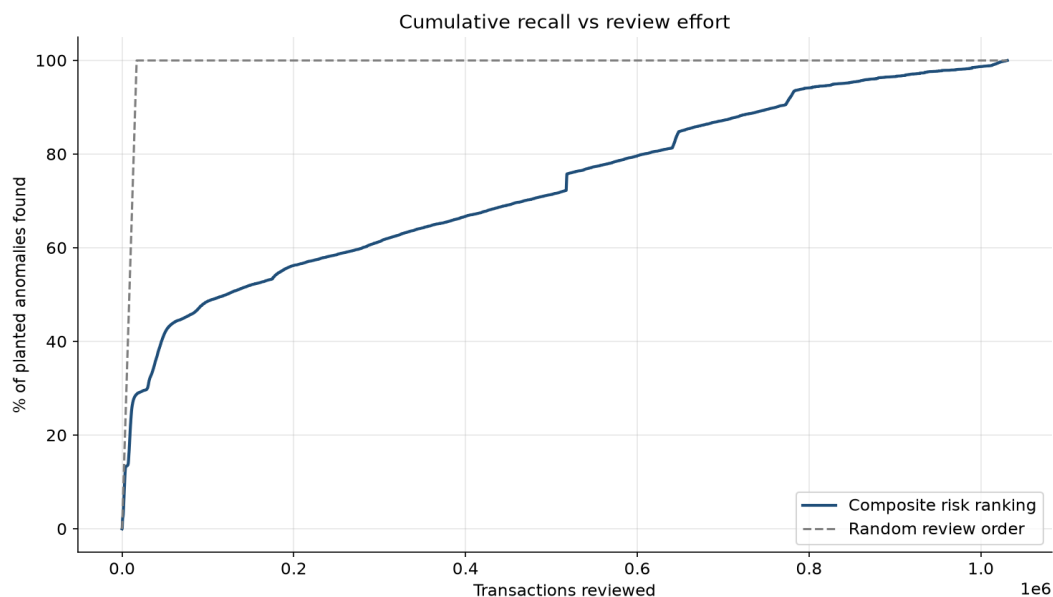


Figure 3: Reviewing the top 500 items surfaces 1.5% of the known synthetic exceptions, against a 818.5% share expected from a random 500-item selection.

6. Schedule of Items for Review

The following 25 transactions carry the highest composite risk scores in the scored population and are recommended for the procedures indicated.

#	Txn ID	Date	Ctry	Amt (£)	Risk	Signals	Procedure
1	00007200	12-03 12:31	UK	1080	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
2	00032664	12-15 15:48	UK	960	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
3	00413870	10-29 12:09	UK	960	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
4	00924000	11-02 11:24	UK	960	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
5	00231315	06-14 11:48	UK	640	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
6	00803007	08-24 10:52	UK	640	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
7	00615136	03-17 10:17	UK	1980	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
8	00670452	05-06 13:03	UK	640	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
9	00409934	10-27 09:31	UK	780	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
10	00575841	02-04 11:55	UK	780	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
11	01005594	12-01 10:07	UK	780	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
12	00300267	08-15 10:33	UK	480	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
13	00583704	02-15 10:17	UK	480	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
14	00021612	12-09 16:17	UK	480	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
15	00796618	08-18 09:39	UK	480	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
16	00618464	03-20 10:37	UK	540	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
17	00369015	10-06 11:43	UK	480	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
18	00369016	10-06 11:43	UK	480	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
19	00476376	11-24 09:29	UK	390	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
20	00145524	04-01 10:36	UK	390	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
21	00813468	09-01 17:19	Eir	540	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
22	00615623	03-17 15:41	UK	390	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
23	00427395	11-04 15:49	UK	3660	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
24	00710440	06-08 11:43	UK	390	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.
25	00944263	11-10 11:55	UK	1790	1.00	ROUND_NUMBER, LA..	Obtain the approval matrix; confirm the authorisation limit and whether the item was split to avoid it.

Prepared by	_____	Date	_____
Reviewed by	_____	Date	_____